$F = -\frac{dV}{ds}$

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$$\Rightarrow \frac{P_{total,suc}}{\rho g} = \frac{P_s}{\rho g} + \frac{V^2}{2g} = \frac{P_{atm}}{\rho g}$$

Flow rate $Q \propto n$ — $n \rightarrow$ RPM of pump

Pumphead $H \propto n^2$ — $D \rightarrow$ diameter of pump impeller.

Pump power $P \propto n^3$

Pump laws - $\frac{Q}{ND^3}$ $\frac{H}{N^2D^2}$ $\frac{P}{\rho N^3 D^5}$ → Dimensionless

Q: $\eta = 80\%$

$N = 2328$, $D = 200$ mm

$$\frac{Q_1}{N_1 D_1^3} = \frac{Q_2}{N_2 D_2^3}$$

- Conservation Equations

1> Conservation of mass: For a system (defined by a specific amount of matter, M, we have chosen) we have the sample results that M = constant. To express this law as a rate equation, we write

$$\left.\frac{dM}{dt}\right|_{system} = 0$$

where $M_{system} = \int_{sys} dM = \int_{sys} \rho \, dV$

2> Newton's 2nd law: For a system moving relative to an inertial frame of reference, force acting on the system is equal to time rate of change of linear momentum of the system.

$$\vec{F} = \left.\frac{d\vec{P}}{dt}\right|_{system},$$

where $\vec{P} = \int \vec{V} \, dm = \int \vec{V} \rho \, dV$

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3) Angular momentum: The angular momentum principle for a system states that the rate of change of angular momentum is equal to the sum of all torques acting on the system

$$\vec{\tau} = \left.\frac{d\vec{L}}{dt}\right|_{sys}$$

where, $\vec{L}_{sys} = \int \vec{r} \times \vec{v}\, dm = \int \vec{r} \times \vec{v}\, \rho\, d\forall$

Torque can be produced by surface and body forces (here gravity) and also by shafts that the system boundary

$$\vec{\tau} = \vec{r} \times \vec{F_s} + \int \vec{r} \times \vec{g}\, dm + \cdots$$

4) 1st law of thermodynamics: It is a statement of conservation of energy for a system.

$$\delta Q - \delta W = dE$$

In rate form..

$$\dot{Q} - \dot{W} = \frac{dE}{dt}$$

where the total energy is given by

$$E = \int e\, dm = \int e \rho\, d\forall$$

and, ~~$e = KE$~~

$$e = KE + PE + U + \text{Pressure energy}$$

5) 2nd law of Thermodynamics: If an amount of heat δQ is transferred to a system at temp 'T', the 2nd law states that the change in entropy dS

$$dS \geq \frac{\delta Q}{T}$$

on a rate basis.

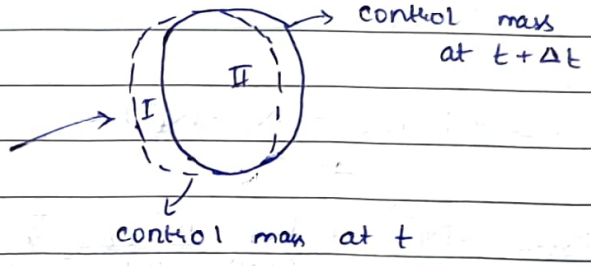
$$\frac{dS}{dT} \geq \frac{1}{T}\dot{Q}$$

where the total entropy is given by

$$S = \int_{sys} s\, dm = \int_{sys} s \rho\, dV$$

- Integral forms of conservation eqn.

★ Reynolds Transport Theorem



Time rate of change of any property for a system = Time rate of change of property within a control volume + the next rate of ~~the~~ efflux of the property from the control volume.

$$\frac{DN}{Dt} = \frac{\partial}{\partial t} \int_{CV} \rho \eta\, dV + \int_{CS} \rho \eta (\vec{v} \cdot \hat{n})\, dA$$

N → extensive property

η → ~~intensi~~ intensive property.

$$N_{sys} = \int \eta\, dm = \int \eta \rho\, dV$$

If $N = M$, $\eta = 1$

$N = \vec{p}$, $\eta = \vec{v}$

$N = H\vec{L}$, $\eta = \vec{r} \times \vec{v}$

$N = E$, $\eta = e$

$N = S$, $\eta = s$

Objective:

To get $\left.\frac{dN}{dt}\right|_{system}$

$$\left.\frac{dN}{dt}\right|_{sys} = \lim_{\Delta t \to 0} \frac{N_{t+\Delta t} - N_t}{\Delta t}$$



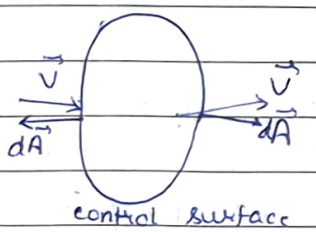
$$N_t = N_t^{I} + N_t^{II}$$

$$N_{t+\Delta t} = N_{t+\Delta t}^{II} + N_{t+\Delta t}^{III}$$

$$\left.\frac{dN}{dt}\right|_{sys} = \lim_{\Delta t \to 0} \frac{\left(N_{t+\Delta t}^{II} + N_{t+\Delta t}^{III}\right) - \left(N_t^{I} + N_t^{II}\right)}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\left(N_{t+\Delta t}^{II} - N_t^{II}\right)}{\Delta t} + \lim_{\Delta t \to 0} \frac{N_{t+\Delta t}^{III}}{\Delta t} - \lim_{\Delta t \to 0} \frac{N_t^{I}}{\Delta t}$$

$$= \left.\frac{\partial N}{\partial t}\right|_{cv} + \text{rate of (outflow - inflow)}$$



Volume flow rate

$= \vec{v} \cdot d\vec{A}$

$= (\vec{v} \cdot \hat{n})\, dA$

$(\vec{v} \cdot \hat{n})\, dA > 0 \rightarrow$ outflow

$< 0 \rightarrow$ inflow

Net efflux $= \int_{cs} \rho \eta\, (\vec{v} \cdot \hat{n})\, dA$

$$\left.\frac{dN}{dt}\right|_{sys} = \left.\frac{\partial N}{\partial t}\right|_{cv} + \int_{cs} \rho \eta\, (\vec{v} \cdot \hat{n})\, dA$$

$$= \int_{cv} \rho \eta\, d\forall + \int_{cs} \rho \eta\, (\vec{v} \cdot \hat{n})\, dA$$

1) Mass conservation on RTT $(N = m,\ \eta = 1)$

$$\left.\frac{dm}{dt}\right|_{sys} = \frac{\partial}{\partial t}\int \rho \, d\forall + \int \rho (\vec{v}\cdot\hat{n})\, dA = 0 \quad \text{(as mass conserved)}$$

Special cases:

- Non deformable CV

then eqn becomes

$$\int_{CV} \frac{\partial \rho}{\partial t}\, d\forall + \int \rho (\vec{v}\cdot\hat{n})\, dA = 0$$

Applying divergence theorem,

$$\int \rho (\vec{v}\cdot\hat{n})\, dA = \int_{CV} \nabla\cdot(\rho\vec{v})\, d\forall$$

Then. $$\int \left[\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\vec{v})\right] d\forall = 0$$

Since steady state, $\frac{\partial \rho}{\partial t} = 0$

$$\Rightarrow \int_{CV} \left[\nabla\cdot(\rho\vec{v})\right] d\forall = 0$$

2) Momentum conservation

$N = \vec{p}$, $\eta = \vec{v}$

$$\frac{d(m\vec{v})}{dt} = \frac{\partial}{\partial t}\int \vec{v}\rho \, d\forall + \int \vec{v}\rho(\vec{v}_r\cdot\hat{n})\, dA = \Sigma\vec{F}$$

It is convinient to express fluid velocities in a frame of reference of coordinates attached to control volume.

Substituting $\vec{p} = m\vec{v}_r$

$$\frac{d(m\vec{v}_r)}{dt} = \frac{\partial}{\partial t}\int_{CV} \vec{v}_r \rho \, d\forall + \int_{CS} \vec{v}_r \rho(\vec{v}_r\cdot\hat{n})\, dA$$

$$\frac{d(m\vec{V}_r)}{dt} = m\frac{d(\vec{V}_r)}{dt}$$

$$= m\frac{d(\vec{V}-\vec{V}_c)}{dt}$$

$$= m\frac{d\vec{V}}{dt} - \frac{m\vec{V}_c}{dt}$$

$$= m\frac{d\vec{V}}{dt}\Big|_{cms} - m\vec{a}_c$$

$\vec{a}_c = \frac{d\vec{V}_c}{dt}$ → rectilinear acceleration of the CV (in a fixed coordinate sys), may or may not be a function of time.

$$m\frac{d\vec{V}_r}{dt} = \Sigma\vec{F} - m\vec{a}_c$$

At steady state,

$$\Sigma\vec{F} - m\vec{a}_c = \int \vec{V}_r \rho(\vec{V}_r \cdot \hat{n})\, dA$$

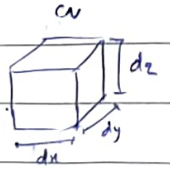
If CV moves at const rectilinear vel, $\vec{a}_c = 0$

$$\Sigma\vec{F} = \int \vec{V}_r \rho(\vec{V}_r \cdot \hat{n})\, dA$$

- Differential analysis of fluid motion.

1) Conservation of mass: The density and velocity fields are related through the following mass conservation eqn.

$$\frac{\partial}{\partial t}\int_{CV} \rho\, d\forall + \int_{CS} \rho\, \vec{V}\cdot d\vec{A} = 0$$



$$\frac{\partial}{\partial t}\int_{CV} \rho\, d\forall = \frac{\partial \rho}{\partial t}\, dx\, dy\, dz$$

In this intinitesimal small volume,

velocity = $u\hat{i} + v\hat{j} + w\hat{k}$

To evaluate the properties at each of six faces of CV, use taylor expansion about point O (origin).

At right face

$$\rho\Big|_{x+dx/2} = \rho + \left(\frac{\partial \rho}{\partial x}\right)\frac{dx}{2} + \left(\frac{\partial^2 \rho}{\partial x^2}\right)\frac{1}{2!}\left(\frac{dx}{2}\right)^2 + \left(\frac{\partial^3 \rho}{\partial x^3}\right)\frac{1}{3!}\left(\frac{dx}{2}\right)^3 + \cdots$$

Neglecting higher order,

$$\rho\Big|_{x+\frac{dx}{2}} = \rho + \left(\frac{\partial \rho}{\partial x}\right)\frac{dx}{2}$$

$$u\Big|_{x+\frac{dx}{2}} = u + \left(\frac{\partial u}{\partial x}\right)\frac{dx}{2}$$

At ~~text~~ left face,

$$\rho\Big|_{x-\frac{dx}{2}} = \left(\frac{\partial \rho}{\partial x}\right)\rho - \left(\frac{\partial \rho}{\partial x}\right)\frac{dx}{2}$$

Left surface (-)

$$= -\left[\rho - \left(\frac{\partial \rho}{\partial x}\right)\frac{dx}{2}\right]\left[u - \left(\frac{\partial u}{\partial x}\right)\frac{dx}{2}\right]dy\,dz = -\rho u\,dy\,dz + \frac{1}{2}\left[u\left(\frac{\partial \rho}{\partial x}\right) + \rho\left(\frac{\partial u}{\partial x}\right)\right]dx\,dy\,dz$$

Right surface (+)

$$= \rho u\,dy\,dz + \frac{1}{2}\left[u\left(\frac{\partial \rho}{\partial x}\right) + \rho\left(\frac{\partial u}{\partial x}\right)\right]dx\,dy\,dz$$

Bottom surface (-)

$$= -\rho v\,dx\,dz + \frac{1}{2}\left[v\left(\frac{\partial \rho}{\partial y}\right) + \rho\left(\frac{\partial v}{\partial y}\right)\right]dx\,dy\,dz$$

Top (+)

$$= \rho v \, dx \, dz + \frac{1}{2}\left[v\left(\frac{\partial \rho}{\partial y}\right) + \rho\left(\frac{\partial v}{\partial y}\right)\right] dx \, dy \, dz$$

Back (-)

$$= -\rho w \, dy \, dx + \frac{1}{2}\left[w\left(\frac{\partial \rho}{\partial z}\right) + \rho\left(\frac{\partial w}{\partial z}\right)\right] dx \, dy \, dz$$

Front (+)

$$= +\rho w \, dy \, dx + \frac{1}{2}\left[w\left(\frac{\partial \rho}{\partial z}\right) + \rho\left(\frac{\partial w}{\partial z}\right)\right] dx \, dy \, dz$$

Now,

$$\iint_{CS} \rho \vec{v} \, d\vec{A}$$

$$= \left[\left(u\frac{\partial \rho}{\partial x} + \rho\frac{\partial u}{\partial x}\right) + \left(v\frac{\partial \rho}{\partial y} + \rho\frac{\partial v}{\partial y}\right) + \left(w\frac{\partial \rho}{\partial z} + \rho\frac{\partial w}{\partial z}\right)\right] dx\,dy\,dz$$

$$= \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w)$$

$$= \nabla(\rho \vec{v})$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \quad \text{— (1)}$$

This eqn is known as continuity eqn.

Case I: Incompressible fluid

$$\rho \neq f(t, x)$$

Then, $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$, or $\nabla \cdot v = 0$ — (2)

Case II : Steady state flow

In this case properties are time independent.

$$\nabla \cdot (\rho \vec{v}) = 0$$

$$\Rightarrow \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

Case III - Steady state incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Q: For 2D incompressible flow in xy plane, $u = Ax$. Determine the possible y component. How many y-components are possible.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

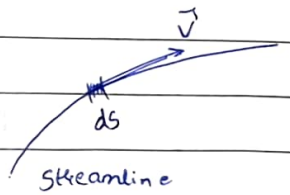
$$\Rightarrow A + \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow v = -Ay + C \quad , \; C \in f(x,t)$$

$$C = f(x,t)$$

- Stream functions

Streamline eqn : $\frac{dy}{dx} = \frac{v}{u}$



$$\vec{v} \times d\vec{s} = 0$$

$$\vec{v} = u\hat{i} + v\hat{j} + z\hat{k}$$

$$d\vec{s} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u & v & z \\ dx & dy & dz \end{vmatrix} = 0 \quad \Rightarrow \quad \begin{aligned} u\,dy - v\,dx &= 0 \\ u\,dz - w\,dx &= 0 \\ v\,dz - w\,dy &= 0 \end{aligned}$$

Hence, $\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$

Stream fn : $\psi(x, y, z, t)$

For 2D flow

$\psi(x, y, t)$, velocity components:

$u(x, y, t)$, $v(x, y, t)$

Then $u = \frac{\partial \psi}{\partial y}$, $v = -\frac{\partial \psi}{\partial x}$

it should satisfy the eqn.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} = 0$$

$$\frac{dx}{u} = \frac{dy}{v}$$

$$\Rightarrow v\,dx - u\,dy = 0$$

$$\Rightarrow -\frac{\partial \psi}{\partial x}\,dx - \frac{\partial \psi}{\partial y}\,dy = 0$$

$$\Rightarrow \frac{\partial \psi}{\partial x}\,dx + \frac{\partial \psi}{\partial y}\,dy = 0$$

$$\Rightarrow d\psi = \frac{\partial \psi}{\partial x}\,dx + \frac{\partial \psi}{\partial y}\,dy$$

comparing the eqn.

$$\underline{d\psi = 0}$$

$\Rightarrow$ ψ is const along a streamline.